Agrannya Singh

+91 9508901080 | singh.agrannya@gmail.com | agrannya | Agrannya-Singh

Education

- **Vellore Institute of Technology** | *B.Tech in CSE (CGPA: 9.05/10)* — *Vellore, TN | Aug 2023 – Jun 2027*
- **Shivam Convent, Patna** | *Class XII, CBSE (94%)* — *Patna, Bihar | Apr 2023*
- **Delhi Public School, Patna** | *Class X, CBSE (97%)* — *Patna, Bihar | Apr 2021*

Experience

- **Novustrana** | *Full Stack Intern (Hybrid)* — *Hyderabad | Mar 2026 – Apr 2026*
  - Architected an Agentic Orchestration System for automated employability scoring by integrating **Firebase Cloud Functions** with **Gemini 2.5-Flash**, developing a POC tested over **100+ user accounts**.
  - Engineered an OSINT Intelligence Check Engine utilizing **Google Search grounding** to verify resume claims, producing a dynamic **Trust Matrix** that identifies digital footprint inconsistencies.
  - Optimized backend reliability by developing **Express.js** API routes with **JWT authentication** and streamlining retrieval via **Firestore composite indexing** and **PITR**.
- **SambalPay Fintech Solutions** | *Software Engineering Intern* — *Gurgaon (Remote) | Feb 2026 – Mar 2026*
  - Architected and Containerized the Core Banking Backend by deploying **Apache Fineract** via **Docker Compose** on **GCP** with optimized 12GB memory allocation to ensure system stability.
- **Mokshapay** | *Founding Intern – Full Stack Engineer* — *Remote | Oct 2025 – Jan 2026*
  - Developed a fintech platform using **Next.js 15** and **Tailwind**; managed 50+ releases via automated CI/CD pipelines (**Cloud Build**, **Docker**) deployed to Firebase App Hosting.
  - Configured production GCP infrastructure via **Terraform**, establishing a compliant VPC with Cloud NAT and reserved static IPs for third-party KYC vendor integration.
  - Architected a backend using **PostgreSQL** for transactional Loan Management (LMS) and **Cloud Firestore** for user profiles, securing sensitive data with fine-grained Security Rules.
  - Built a GenAI-powered helpdesk using **Vertex AI Agent Builder**; integrated **Firebase and Figma MCP servers** to accelerate debugging and development workflows.
- **Labmentex** | *Python Intern* — *Remote | Aug 2025 – Sep 2025*
  - Engineered a custom EDA engine using **Python** and **Flask** to process NASA DONKI datasets including Coronal Mass Ejection (CME), Solar Flare (FLR), and Geomagnetic Storm (GST).
  - Developed a feature selection algorithm utilizing **Pearson Correlation** matrices to identify and filter highly correlated features, effectively reducing dataset noise for downstream analysis.

Technical Skills

- **Languages:** TypeScript, Python, JavaScript, HTML/CSS, SQL
- **Frameworks/Libraries:** Next.js, React, Node.js, FastAPI, Flask, SQLAlchemy, Genkit, Firebase Auth
- **AI/ML:** scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, Transformers
- **Cloud & DevOps:** Docker, AWS (EC2, S3, IAM, **Lambda**), GCP (Cloud Build, Artifact Registry, **Cloud Functions**), Azure Web Apps, GitHub Actions, Firebase App Hosting, Render
- **Databases:** MongoDB Atlas, PostgreSQL, Redis, Pinecone, SQLite, Cloud Firestore

Projects

- **TuneTrace - Music Discovery App** | Live | — *May 2025 – Present*
  Next.js, FastAPI, PostgreSQL, Redis, Azure Web Services
  - Engineered a FastAPI/PostgreSQL microservice on Azure, reducing API latency by **40%** and query speeds to **<200ms** using a dual-layer Redis caching strategy.
  - Built a hybrid Scikit-Learn recommendation engine using **TF-IDF vectorization** to process song metadata, entirely eliminating "cold starts" for new users.
  - Maximized discovery and prevented "filter bubbles" by applying exponential recency decay and algorithmically allocating **40%** of recommendations to diversity sampling.
  - Automated serverless ETL pipelines via GitHub Actions and the YouTube Data API v3 to seed weekly trending data, ensuring a constantly updated catalog with zero manual maintenance.
- **ScreenScout - Semantic Vector Search Engine** | HF | — *Jun 2025 – Present*
  Next.js, FastAPI, Sentence Transformers, Pinecone, SQLite, Supabase, Azure Web Apps, GitHub Actions
  - Architected a scalable 3-tier semantic search engine deployed on Azure Web Apps, indexing **30,000+** movie records to facilitate high-performance content discovery.
  - Integrated **Pinecone Vector Database** to manage high-dimensional embeddings generated by the **all-MiniLM-L6-v2** Sentence Transformer model, enabling **sub-100ms** approximate nearest neighbor (ANN) retrieval.
  - Implemented a hybrid architecture using **SQLite** for ultra-fast vector lookups and **Supabase** for analytics, delivering a total API response time (TTFB) of **100ms**, with seamless CI/CD integration via GitHub Actions to Azure Web

Apps.

- **Othello Dojo** Live | | k *Jul 2025 – Present*
  Next.js, Node.js, Tailwind CSS, Python, MongoDB, Flask, TensorFlow
  - Led a team of 5 to develop an autonomous Reversi agent, combining traditional **Minimax** algorithms with modern deep learning techniques for competitive gameplay.
  - Engineered a custom **Monte Carlo Tree Search (MCTS)** pipeline to generate 10,000+ synthetic training states, training a **ResNet-based policy-network (OthelloNetV3)** that achieved a **70% win rate**.
  - Implemented the core **Othello game logic** with complete rule validation, move generation, piece-flipping mechanics, and scoring systems.
- **AI-Powered Space Weather Intelligence Dashboard** Live | *Jul 2024 – Present*
  Next.js, TypeScript, Genkit, Docker
  - Constructed a full-stack dashboard to visualize **NASA space weather datasets** with over **50,000 records** in real-time; created a custom **EDA engine** to automate chart generation.
  - Leveraged Google's **Gemini AI via Genkit** to generate contextual, **EDA-aware summaries**, making complex scientific data accessible to a broader audience.

Certifications

- **Generative AI using IBM Watson X** *IBM* Link
- **Machine Learning for Data Science Projects** *IBM SkillsBuild* Link
- **Using Atlas Vector Search for RAG Applications** *MongoDB University* Link
- **Cloud Computing Fundamentals** *IBM SkillsBuild* Link
- **Cybersecurity Fundamentals** *IBM SkillsBuild* Link
- **DevOps and Site Reliability Engineering** *The Linux Foundation* Link